

Lab Assignment # 09

Course Title : AI Assistant Coding

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Batch No. :48

Lab 9: Documentation Generation – Automatic Documentation and Code Comments

Task 1: Basic Docstring Generation

Scenario: You are developing a utility function that processes numerical lists and must be Properly documented for future maintenance.

Prompt: Write a Python function to return the sum of even numbers and sum of odd numbers in a given list. Generate a Google-style Python docstring for the following function. The docstring should include: A clear description of the function's purpose, Args section with parameter names and types, Returns section with types and descriptions

Expected Output:

- Python function with manual Google-style docstring
- AI-generated docstring for the same function
- Comparison explaining differences between manual and AI-generated documentation
- Improved understanding of AI-generated function-level documentation.

Output Screenshot:

```
LAB_09.PY > ...
1 #Lab 9: Documentation Generation - Automatic Documentation and Code Comments
2 #task description #1
3 #Write a Python function to return the sum of even numbers and sum of odd numbers in a given list
4 #The docstring should include: A clear description of the function's purpose, Args section with
5 # names and types, Returns section with types and descriptions
6 def sum_even_odd(numbers):
7     """
8     Calculate the sum of even and odd numbers in a given list.
9
10    Args:
11        numbers (list): A list of integers to be processed.
12    Returns:
13        tuple: A tuple containing two elements:
14            - sum_even (int): The sum of all even numbers in the list.
15            - sum_odd (int): The sum of all odd numbers in the list.
16    """
17    sum_even = sum(num for num in numbers if num % 2 == 0)
18    sum_odd = sum(num for num in numbers if num % 2 != 0)
19    return sum_even, sum_odd
20
21 #Example usage:
22 numbers = [1, 2, 3, 4, 5, 6]
23 even_sum, odd_sum = sum_even_odd(numbers)
24 print(f"Sum of even numbers: {even_sum}") # Output: Sum of even numbers: 12
25 print(f"Sum of odd numbers: {odd_sum}") # Output: Sum of odd numbers: 9
26
27 #task description #2
28 #You are developing a student management module that must be easy to understand for new developers.
29 #Requirements: Attributes: name, roll_no, hostel_status, Methods: fee_update() and display_details()
30 class Student:
31     def __init__(self, name: str, roll_no: int, hostel_status: bool):
32         # Initialize the Student object with name, roll number, and hostel status
33         self.name = name # Store the student's name
34         self.roll_no = roll_no # Store the student's roll number
35         self.hostel_status = hostel_status # Store whether the student is in a hostel or not
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\Sai Pallavi\Desktop\AI(LAB)> & "C:\Users\Sai Pallavi\AppData\Local\Programs\Python\Python314\python.exe" "C:/Users/Sai Pallavi/Desktop/AI(LAB)/LAB_09.PY"
Sum of even numbers: 12
Sum of odd numbers: 9
```

Task 2: Automatic Inline Comments

Scenario: You are developing a student management module that must be easy to understand for new developers.

Prompt: You are developing a student management module that must be easy to understand for new developers. Requirements: Attributes: name, roll_no, hostel_status, Methods: fee_update() and display_details(), Manually write inline comments for each line or logical block, Use an AI-assisted tool to automatically add inline comments, Compare manual comments with AI-generated comments, Identify missing, redundant, or incorrect AI comments

Expected Output:

- Python class with manually written inline comments
- AI-generated inline comments added to the same code
- Comparative analysis of manual vs AI comments
- Critical discussion on strengths and limitations of AI-generated comments

Output Screenshot:

```
LAB_09.PY > ...
25
26 #task description #2
27 #You are developing a student management module that must be easy to understand for new developers
28 #Requirements:Attributes: name, roll_no, hostel_status, Methods: fee_update() and display_details()
29 class Student:
30     def __init__(self, name: str, roll_no: int, hostel_status: bool):
31         # Initialize the Student object with name, roll number, and hostel status
32         self.name = name # Store the student's name
33         self.roll_no = roll_no # Store the student's roll number
34         self.hostel_status = hostel_status # Store whether the student is in a hostel or not
35     def fee_update(self, new_fee: float):
36         # Update the fee for the student (this is a placeholder method)
37         pass # This method can be implemented to update the fee based on certain conditions
38     def display_details(self):
39         # Display the details of the student
40         print(f"Name: {self.name}") # Print the student's name
41         print(f"Roll No: {self.roll_no}") # Print the student's roll number
42         print(f"Hostel Status: {'Yes' if self.hostel_status else 'No'}") # Print whether the student is in a hostel or not
43 #Example usage:
44 student1 = Student("Alice", 101, True)
45 student1.display_details() # Output: Name: Alice, Roll No: 101, Hostel Status: Yes
46
47 #task description #3
48 #You are building a small calculator module that will be shared across multiple projects and requires structured documentation.
49 #Write a Python script containing 3-4 functions (e.g., add, subtract, multiply, divide),Manually write NumPy Style
50 #Individual function-level docstrings,Evaluate documentation structure, accuracy, and readability
51 """Calculator Module
52 This module provides basic arithmetic operations such as addition, subtraction, multiplication, and division.
53 Functions:
54 - add(a, b): Returns the sum of a and b.
55 - subtract(a, b): Returns the difference of a and b.
56 - multiply(a, b): Returns the product of a and b.
57 - divide(a, b): Returns the quotient of a and b. Raises ValueError if b is zero.
58 """
59
60 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE
PS C:\Users\Sai Pallavi\Desktop\AI(LAB)> & "C:\Users\Sai Pallavi\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Sai Pallavi/Desktop/AI(LAB)/LAB_09.PY"
Name: Alice
Roll No: 101
Hostel Status: Yes
```

Task 3: Module-Level and Function-Level Documentation

Scenario: You are building a small calculator module that will be shared across multiple projects and requires structured documentation.

Prompt: You are building a small calculator module that will be shared across multiple projects and requires structured documentation. Write a Python script containing 3–4 functions (e.g., add, subtract, multiply, divide), Manually write NumPy Style docstrings for each function, Use AI assistance to generate: A module-level docstring, Individual function-level docstrings, Evaluate documentation structure, accuracy, and readability

Expected Output

- Python script with manual NumPy-style docstrings
- AI-generated module-level and function-level documentation
- Comparison between AI-generated and manual documentation
- Clear understanding of structured documentation for multi-function script.

Output Screenshot:

```
LAB_09.PY > ...
47 #task description #3
48 #You are building a small calculator module that will be shared across multiple projects and
49 #Write a Python script containing 3-4 functions (e.g., add, subtract, multiply, divide),Manual
50 #Individual function-level docstrings,Evaluate documentation structure, accuracy, and readabi
51 """
52 Calculator Module
53 This module provides basic arithmetic operations such as addition,
54 subtraction, multiplication, and division.
55 """
56
57 def add(a, b):
58     """Add two numbers."""
59     return a + b
60
61
62 def subtract(a, b):
63     """Subtract the second number from the first."""
64     return a - b
65
66
67 def multiply(a, b):
68     """Multiply two numbers."""
69     return a * b
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71
72 def divide(a, b):
73     """Divide the first number by the second."""
74     if b == 0:
75         raise ZeroDivisionError("Cannot divide by zero.")
76     return a / b
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79 # ----- Execute Functions to Get Output -----
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